

Process Notes: Impurity Control and What-If Analysis

Impurity formation in this synthetic process increases when temperature and pressure are both high. Long residence times may improve yield but can also increase degradation in some real systems. The simplified test data only partially captures this tradeoff.

A what-if analysis should vary one parameter at a time initially. Suggested ranges are temperature from 70 to 95 C, pressure from 2.0 to 3.5 bar, and residence time from 30 to 60 minutes.

For optimization, a useful objective is maximize yield_pct while keeping impurity_pct below 8 percent. This creates a constrained optimization problem that is suitable for a future experiment recommendation module.

The AI assistant should not claim that these notes are real plant evidence. They are synthetic notes for testing retrieval, RAG, simulation, and optimization workflows.